

PROVEIT RESEARCH CONTINUATION

Uniform Asymptotics and Zero Bifurcations for Polylogarithmic Families

Harmonic depth, reflected log-gamma moments,
Lerch derivatives, and a weight-nine identity candidate

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The statements below distinguish ordinary analytic proofs, proofs with finite exact rational certificates, and numerical conjectures. The accompanying programs reconstruct their certificates from defining formulas. No claim of proof-assistant formalization or numerical period independence is made.

Abstract

This continuation addresses parameter ranges and zero phenomena left unresolved by the consolidated ProveIt polylogarithm manuscript. For alternating elementary-harmonic sums we prove a saddle expansion with a relative remainder uniform in the harmonic depth, the shift, and the ratio of outer exponent to depth. It gives one formula across the proportional-depth regime and the transition near $p = h \log h$. For reflected log-gamma moments we establish a unique saddle for every positive ratio of the two exponents, with an exact rational certificate for the sole compact-interval inequality needed in the proof, and derive both a joint Laplace expansion and a separate critical-window expansion. For the Stieltjes–Lerch derivative family we determine how an endpoint multiple zero unfolds: at first derivative order there are only one or two positive zeros for sufficiently small positive Lerch parameter, according to the parity of the index. Fresh rational sign certificates then force an interior multiple-root transition at indices three and four. Finally, a new weight-nine candidate reduces S_8 to four single-color Gaussian double polylogarithms and explicit products. Its very small rigorously enclosed residual is kept separate from a proof of equality. The article supplies full derivations, a source audit, reproducible code, and a focused further research programme.

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1 Scope, source status, and principal contributions

The starting point is the consolidated manuscript *Polylogarithms and their Arithmetic Bridges*, at the fixed revision shown on the title page [1]. Its present form already includes the octahedral certificate for S_4 , all-weight Gaussian parity relations, complementary-depth transport, conductor normal forms, positive-kernel zero bounds, and several distinct asymptotic regimes. Those are inputs to this continuation. In particular, the older reports calling S_4 conjectural are superseded by the current source.

The work here follows four concrete questions that remain meaningful after those integrations. Can one control the harmonic-depth saddle while its location moves through the transition range? What happens to reflected moments when both exponents grow? Does the eventual Stieltjes–Lerch zero count extend to every small derivative order at positive parameter? Which next Gaussian Euler sum has a useful, precisely testable reduction?

Problem	Result in this article	Evidence type
Harmonic depth with moving p/h	All-orders relative expansion, uniform for $p \rightarrow \infty$, $h \geq 1$, and $a > 0$	Analytic proof
Two growing reflected exponents	Unique saddle, joint expansion, and critical-window expansion	Analytic proof; one finite exact inequality certificate
Small-parameter Lerch zeros	Puiseux unfolding and exact small-parameter counts; interior multiple-root transitions	Analytic proof; endpoint sign certificates
Mixed Gaussian weight nine	A frozen S_8 relation with independent evaluation and rational residual enclosure	Conjecture with certified numerical evidence

1.1 What is new relative to the source

The harmonic theorem supplies the expanding-range estimate explicitly requested at the end of the source’s proportional-depth section. It retains the shift in the saddle, so the bound does not silently lose uniformity when p/h grows. The reflected-moment work addresses simultaneous growth, which is excluded from the source’s fixed-reflected-exponent remainder theorem. The Lerch analysis proves a specific obstruction to extending eventual saturation to every strictly positive parameter at low derivative order. The S_8 candidate advances the sequence of mixed reductions beyond the source’s S_6 conjecture.

Laplace’s method, gamma concentration, analytic root perturbation, and interval arithmetic are established methods [2, 7]. The contributions claimed here are the stated family-specific theorems, parameter ranges, coefficients, and certificates. A bounded literature check cannot establish worldwide priority; no such priority claim is needed for the mathematical statements or their use in ProveIt. The classical work on log-gamma moments and on cyclotomic polylogarithm relations remains the appropriate background [8, 10, 9].

1.2 Conventions

All real logarithms have positive arguments. We use

$$\mathrm{Li}_{s_1, \dots, s_d}(z_1, \dots, z_d) = \sum_{n_1 > \dots > n_d \geq 1} \frac{z_1^{n_1} \dots z_d^{n_d}}{n_1^{s_1} \dots n_d^{s_d}}.$$

Thus the colors are attached to decreasing summation indices, with the outer index first. Write

$$\beta(s) = \sum_{n \geq 0} \frac{(-1)^n}{(2n+1)^s}, \quad G = \beta(2), \quad g_{a,b} = \Im \operatorname{Li}_{a,b}(i, 1).$$

The Gaussian values used in the new candidate are absolutely convergent. In the asymptotic sections, an assertion uniform in parameters means that the remainder constant is independent of those parameters throughout the stated set. It does not mean that the underlying asymptotic series converges.

2 One saddle expansion throughout the harmonic-depth range

The source manuscript proves separate expansions when p/h lies in a fixed compact interval, when $p/h = \log h + c$, and when p is fixed. It explicitly leaves a uniform connection between the first two regimes open [1]. We supply that connection and a larger parameter range. The central point is to keep the shift a in the saddle equation and to use p as the expansion parameter.

2.1 The sum and its positive integral

For an integer $h \geq 1$ put

$$e_h(n) = [u^h] \prod_{j=1}^n (1 + u/j), \quad T_{p,h}(a) = \sum_{n=0}^{\infty} \frac{(-1)^n e_h(n)}{(n+a)^p}, \quad p, a > 0.$$

The ordinary sum converges for every $p > 0$; for $p > 1$ it converges absolutely. To see the former claim directly, the elementary-symmetric function bound $e_h(n) \leq H_n^h/h!$ gives decay of the terms, and $e_h(n+1) - e_h(n) = e_{h-1}(n)/(n+1)$ gives eventual decrease of $e_h(n)/(n+a)^p$. For example $e_h(n) \sim (\log n)^h/h!$, as follows by expanding the fixed-degree coefficient in $\exp(\sum_{j \geq 1} (-1)^{j-1} H_n^{(j)} u^j/j)$.

Define the positive normalization

$$R_{p,h}(a) = (-1)^h h! (h+a)^p T_{p,h}(a). \quad (2.1)$$

The generating function

$$\sum_{n \geq 0} e_h(n) z^n = \frac{(-\log(1-z))^h}{h!(1-z)}$$

and the gamma integral give

$$R_{p,h}(a) = \frac{(h+a)^p}{\Gamma(p)} \int_0^{\infty} \frac{y^{p-1} e^{-ay} L(y)^h}{1 + e^{-y}} dy, \quad L(y) = \log(1 + e^{-y}). \quad (2.2)$$

One may insert a radial Abel factor before interchanging sums and integrals, then use dominated convergence and the ordinary convergence just established. This is the source manuscript's positive integral, recalled here to make the continuation self-contained.

It is useful to set

$$H(x) = \log(e^x L(x)), \quad w(x) = -\frac{d}{dx} \log L(x) = \frac{1}{(1 + e^x) L(x)}. \quad (2.3)$$

Then $H' = 1 - w$, $H'' = -w' < 0$, and

$$w_0 := \frac{1}{2 \log 2} < w(x) < 1, \quad w'(x) > 0. \quad (2.4)$$

Indeed, with $t = e^{-x}$, $w = t/((1+t) \log(1+t))$, whose logarithmic derivative with respect to t is $(\log(1+t) - t)/(t(1+t) \log(1+t)) < 0$. The function H extends smoothly to $x = 0$; all of its positive-order derivatives decay exponentially as $x \rightarrow \infty$.

2.2 Exact saddle and main theorem

There is exactly one $x = x(p, h, a) > 0$ satisfying

$$x\{hw(x) + a\} = p. \quad (2.5)$$

The left side has a strictly positive derivative and runs from zero to infinity. In particular,

$$\frac{p}{h+a} < x < \frac{p}{hw_0+a}. \quad (2.6)$$

Define

$$\theta = \frac{hx}{p} = \frac{h}{hw(x) + a}, \quad B = 1 + \theta x w'(x), \quad (2.7)$$

$$\delta = \frac{(h+a)x}{p} - 1, \quad E = p\{\log(1+\delta) - \delta\} + hH(x), \quad (2.8)$$

$$\mathcal{A}(p, h, a) = \frac{e^E}{(1 + e^{-x})\sqrt{B}}. \quad (2.9)$$

All these quantities are real. Both summands in E are nonpositive, which also makes (2.8) a useful form for numerical evaluation.

Theorem 2.1 (Uniform moving-saddle expansion). *For each integer $J \geq 0$ there are bounded, explicitly computable coefficient functions $d_j(x, \theta)$, $0 \leq j \leq J$, with $d_0 = 1$, and constants C_{J, p_J} depending only on J , such that*

$$\left| \frac{R_{p,h}(a)}{\mathcal{A}(p, h, a)} - \sum_{j=0}^J \frac{d_j(x, \theta)}{p^j} \right| \leq \frac{C_J}{p^{J+1}} \quad (2.10)$$

for every real $p \geq p_J$, every integer $h \geq 1$, and every $a > 0$. There is no lower or upper restriction on p/h , and a may also vary. The constants in the error bound are independent of all three parameters. In particular, $R_{p,h}(a) = \mathcal{A}(p, h, a)(1 + O(p^{-1}))$ uniformly.

Here is the first coefficient in a form that only uses derivatives at the single real saddle. Write

$$\begin{aligned} v &= (1 + e^x)^{-1}, & \ell_1 &= -1 + xv, & \ell_2 &= 1 - x^2 v(1 - v), \\ b_3 &= 2 - \theta x^2 w''(x), & b_4 &= -6 - \theta x^3 w'''(x). \end{aligned} \quad (2.11)$$

Then

$$d_1 = \frac{\ell_2 + \ell_1^2}{2B} + \frac{b_3 \ell_1}{2B^2} + \frac{b_4}{8B^2} + \frac{5b_3^2}{24B^3} - \frac{1}{12}. \quad (2.12)$$

The final $-1/12$ is the gamma-normalization correction. Omitting it would fail even in the limiting pure-gamma model.

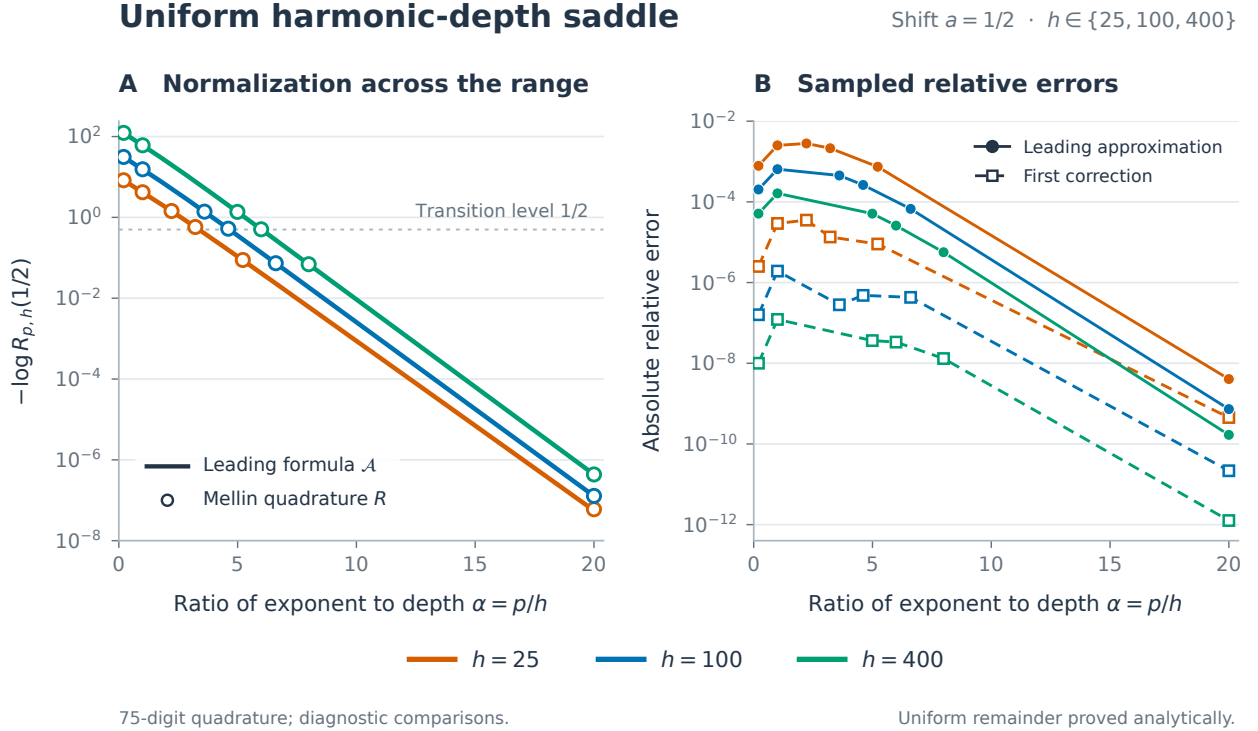


Figure 1: Uniform harmonic-depth approximation at $a = 1/2$, for $h = 25, 100, 400$. Both vertical axes are logarithmic. **(A)** Solid curves show $-\log \mathcal{A}$ from the explicit moving-saddle formula over $0.2 \leq \alpha = p/h \leq 20$; circles show $-\log R$ from 75-digit Mellin quadrature. The horizontal guide at $1/2$ is the transition limit $-\log R_{h \log h, h}(1/2) \rightarrow 1/2$. **(B)** Absolute relative errors of the leading approximation $\mathcal{A}$ and its first correction $\mathcal{A}(1 + d_1/p)$ at the 18 sampled parameters. Connecting segments guide the eye. These are numerical diagnostics; the analytic proof of Theorem 2.1 supplies the uniform remainder estimate.

2.3 A uniform Laplace lemma and proof

Put $y = xu$ in (2.2). The phase and amplitude, normalized at their unique saddle $u = 1$, are

$$\phi(u) = \log u - u + 1 + \frac{\theta}{x} \{H(xu) - H(x) - xH'(x)(u - 1)\}, \quad (2.13)$$

$$g_x(u) = \frac{1}{u(1 + e^{-xu})}. \quad (2.14)$$

Thus $\phi(1) = \phi'(1) = 0$, $-\phi''(1) = B$, and the exact integral is

$$R_{p,h}(a) = \frac{(h+a)^p x^p L(x)^h e^{-ax}}{\Gamma(p)} \int_0^\infty e^{p\phi(u)} g_x(u) du. \quad (2.15)$$

The following four bounds hold uniformly in $x > 0$ and $0 \leq \theta \leq 1/w(x)$, a set containing every saddle parameter above.

1. Concavity of H gives the global bound

$$\phi(u) \leq \log u - u + 1 \quad (u > 0). \quad (2.16)$$

2. The curvature is uniformly nondegenerate: $1 \leq B \leq 1 + w_0^{-1} \sup_{x>0} xw'(x) < \infty$.
3. On $1/2 \leq u \leq 3/2$, each fixed derivative of ϕ is uniformly bounded. For $m \geq 2$ the only extra term beyond a derivative of $\log u - u + 1$ is $\theta x^{m-1} H^{(m)}(xu)$. Its boundedness follows from smoothness at zero and exponential decay at infinity. Derivatives of g_x are uniformly bounded on the same interval, while $1/2 \leq g_x(1) \leq 1$.
4. Globally $0 < g_x(u) \leq 1/u$.

For completeness, these bounds suffice for a parameter-uniform Laplace expansion, rather than merely a pointwise use of the method [2, 7]. Choose a fixed $0 < \eta < 1/2$. The contribution of $|u - 1| \geq \eta$ is bounded by

$$\int_{|u-1| \geq \eta} \frac{e^{p(\log u - u + 1)}}{u} du \leq Cp^{-1/2} e^{-c_\eta p}.$$

One direct proof divides the integral by $e^p \Gamma(p)/p^p$ and applies the elementary exponential Markov bound to a gamma random variable of shape p and rate p . The two tail exponents are $1 \pm \eta - 1 - \log(1 \pm \eta) > 0$. This controls the unbounded domain and the singular factor $1/u$ at once.

On $|u - 1| < \eta$, set $u = 1 + z/\sqrt{p}$. Taylor-expand the phase and amplitude to a fixed order. Uniform derivative bounds give uniform Taylor remainders. The inequality (2.16) supplies a fixed Gaussian majorant in this neighborhood. Splitting further at $|z| = p^\varepsilon$, with a sufficiently small fixed $\varepsilon > 0$ for the chosen order, justifies expansion of the exponential with an integrable polynomial-times-Gaussian remainder. Outside that smaller interval the same majorant is smaller than every inverse power of p . Odd powers integrate to zero on the symmetric neighborhood, and extending each polynomial Gaussian integral to the whole real line changes it by an exponentially small quantity. Expanding two additional orders when necessary gives, for every fixed J ,

$$\int_0^\infty e^{p\phi(u)} g_x(u) du = g_x(1) \sqrt{\frac{2\pi}{pB}} \left\{ \sum_{j=0}^J \frac{c_j(x, \theta)}{p^j} + O_J(p^{-J-1}) \right\}. \quad (2.17)$$

Every division here is by a quantity uniformly bounded away from zero. This also proves uniform boundedness of each c_j .

Let $\Gamma^*(p) = \Gamma(p)/(\sqrt{2\pi}p^{p-1/2}e^{-p})$. Substitution of (2.17) into (2.15) gives $\mathcal{A}/\Gamma^*(p)$ times the bracketed expansion. The ordinary positive-real Stirling expansion, with its remainder [3], now proves Theorem 2.1 by multiplication of two finite expansions. The displayed $c_1 = d_1 + 1/12$ is obtained from the Gaussian moments of degrees two, four and six. This proves (2.12) as well.

2.4 A finite coefficient compiler

The proof specifies every coefficient without fitting any data. Let $\phi_m = \phi^{(m)}(1)$ and $g_m = g_x^{(m)}(1)$. For $j \geq 0$ define the polynomial

$$P_j(z) = [t^{2j}] \left(\sum_{r=0}^{2j} \frac{g_r z^r t^r}{r!} \right) \exp \left(\sum_{m=3}^{2j+2} \frac{\phi_m z^m t^{m-2}}{m!} \right). \quad (2.18)$$

With the convention that odd Gaussian moments vanish,

$$c_j = \frac{1}{g_0} \sqrt{\frac{B}{2\pi}} \int_{\mathbb{R}} e^{-Bz^2/2} P_j(z) dz. \quad (2.19)$$

The even moments are $(2r-1)!!B^{-r}$ after normalization. Finally, as a formal power-series prescription,

$$\sum_{j \geq 0} d_j \varepsilon^j = \exp \left(- \sum_{r \geq 1} \frac{B_{2r} \varepsilon^{2r-1}}{2r(2r-1)} \right) \sum_{j \geq 0} c_j \varepsilon^j. \quad (2.20)$$

Only finitely many terms enter each coefficient. The expansion is an asymptotic expansion; no convergence of (2.20) is asserted.

2.5 Recovery of existing regimes and a new uniform connection

For a fixed and $p = \alpha h$ with α in a positive compact interval, the new saddle solves $x(w(x) + a/h) = \alpha$. Let $x_\alpha w(x_\alpha) = \alpha$ be the source saddle. The implicit function theorem gives

$$x = x_\alpha - \frac{ax_\alpha}{h\{w(x_\alpha) + x_\alpha w'(x_\alpha)\}} + O(h^{-2}).$$

The envelope derivative in a/h , or direct substitution, then gives

$$\mathcal{A}(p, h, a) = C(\alpha, a) e^{-hI(\alpha)} (1 + O(h^{-1})), \quad (2.21)$$

where the source functions are

$$I(\alpha) = -\alpha \log(x_\alpha/\alpha) - \alpha - \log L(x_\alpha), \quad C(\alpha, a) = \frac{\sqrt{\alpha}}{\sqrt{\alpha + x_\alpha^2 w'(x_\alpha)}} \frac{e^{a(\alpha - x_\alpha)}}{1 + e^{-x_\alpha}}.$$

The new result retains the actual saddle and therefore does not expand a term whose error would grow with α .

At the transition $p = h(\log h + c)$, a, c in fixed compact sets, write $\alpha = \log h + c$ and $\lambda = e^{-c}/2$. Since $w(x) = 1 - e^{-x}/2 + O(e^{-2x})$, the saddle has $x = \alpha + O(\alpha/h)$, and consequently

$$hH(x) = -\lambda + O(\alpha/h), \quad p\{\log(1 + \delta) - \delta\} = O(\alpha/h), \quad B = 1 + O(\alpha/h).$$

It follows from the uniform theorem that

$$R_{p,h}(a) = e^{-\lambda}\{1 + O(\alpha/h)\}.$$

The source already proves a sharper transition expansion; the new contribution is that the same unexpanded $\mathcal{A}$ has a uniform relative error $O(1/p)$ across this transition and all intervening values of p/h . Including d_1/p improves it to $O(p^{-2})$.

For any fixed $\alpha_0 > 0$ and any function $M(h) \geq \alpha_0$, Theorem 2.1 gives, without an upper growth condition,

$$\sup_{\substack{\alpha_0 \leq p/h \leq M(h) \\ a > 0}} \left| \frac{R_{p,h}(a)}{\mathcal{A}(p, h, a)} - 1 \right| = O(h^{-1}). \quad (2.22)$$

The corrected supremum is $O(h^{-2})$. This explicitly resolves the expanding-range question in the proportional-depth chapter.

At $x \rightarrow \infty$, the phase tends to $\log u - u + 1$, the amplitude tends to $1/u$, and all d_j for $j \geq 1$ tend to zero. Thus the gamma normalization is consistent with first-term dominance when also $he^{-x} \rightarrow 0$, as happens for fixed h . The limit $x \rightarrow \infty$ alone does not imply $R \rightarrow 1$ when the depth varies. Conversely, the theorem also allows $p/h \rightarrow 0$ provided $p \rightarrow \infty$; it makes no claim about a bounded p , where a different endpoint expansion is appropriate.

2.6 Exact monotonicities

The same positive representation gives useful checks on numerical work. If Y is gamma with shape p and rate $h + a$, then

$$R_{p,h}(a) = \mathbb{E} k_h(Y), \quad k_h(y) = \frac{(e^y L(y))^h}{1 + e^{-y}}.$$

The function k_h is strictly increasing in y and strictly decreasing in h . Coupling gamma variables by addition proves that R is strictly increasing in p . Scaling a common gamma variable proves strict decrease in a , and jointly increasing the rate and decreasing the kernel proves strict decrease in the integer h for fixed p, a . Also $0 < R < 1$. These are exact statements, independent of an asymptotic regime.

3 Joint asymptotics for reflected log-gamma moments

Let

$$f(x) = \log \Gamma(x), \quad B(x) = f(1 - x), \quad M_{n,m} = \int_0^1 f(x)^n B(x)^m dx.$$

These mixed integrals are classical; they occur as $T_{n+m,n}$ in [8, Section 5]. The fixed- m expansions in the repository do not give relative estimates when both exponents grow. This section treats two parts of that unresolved regime. First, a global uniqueness theorem gives a full saddle expansion whenever m/n remains in a compact subset of $(0, \infty)$. Second, an entirely analytic transition theorem determines exactly how the leading fixed- m approximation fails near $n/(m+1) = \log m + O(1)$. The first result uses a short, reproducible rational certificate for a special-function inequality. The transition theorem does not depend on that certificate.

All exponents in this section may be positive real numbers; the factorial $n!$ is then written $\Gamma(n+1)$. The familiar inequalities $0 < f(x) \leq -\log x$ on $(0, 1)$ follow from $\psi(1) = -\gamma < 0$, $\psi'(x) > 0$, and

log-convexity of Γ between 1 and 2. The Taylor series and polygamma formulas used below are classical [5, 6].

3.1 A strict monotonicity theorem and one saddle for every ratio

Define positive functions

$$h(x) = \frac{-\psi(x)}{\log \Gamma(x)}, \quad Q(x) = \frac{h(x)}{h(1-x)}, \quad 0 < x < 1. \quad (3.1)$$

Direct differentiation gives

$$\begin{aligned} D(x) := \frac{d}{dx} \log Q(x) &= \frac{-\psi(x)}{f(x)} + \frac{-\psi(1-x)}{f(1-x)} \\ &\quad - \frac{\psi'(x)}{-\psi(x)} - \frac{\psi'(1-x)}{-\psi(1-x)}. \end{aligned} \quad (3.2)$$

Theorem 3.1 (Strict reflected elasticity inequality). *For every $0 < x < 1$, $D(x) > 0$. Consequently Q is a strictly increasing real-analytic bijection from $(0, 1)$ onto $(0, \infty)$, and $Q(1-x) = Q(x)^{-1}$. In particular,*

$$\log \Gamma(x) \log \Gamma(1-x) \leq \frac{\log^2 \pi}{4}, \quad (3.3)$$

with equality exactly at $x = 1/2$.

Proof. The identity $D(1-x) = D(x)$ reduces the problem to $0 < x \leq 1/2$. We first give an analytic proof on $0 < x \leq 1/16$, and then describe the finite rational certificate for $1/16 \leq x \leq 1/2$.

Write $L = -\log x$, $p = -\psi(x)$, $s = -\psi(1-x)$, and $b = B(x)/x$. The positive series

$$B(x) = \gamma x + \sum_{k \geq 2} \frac{\zeta(k)}{k} x^k$$

implies $s/b \geq 1$ and $s \geq \gamma > 1/2$. Concavity of ψ gives $\psi(1+x) \leq -\gamma + \zeta(2)x < 0$ on this interval. Thus $xp = 1 - x\psi(1+x) \geq 1$, and $f(x) \leq L$ gives $xp/f(x) \geq 1/L$. The polygamma series yield

$$x^2\psi'(x) = 1 + x^2\psi'(1+x) \leq 1 + 2x^2, \quad \psi'(1-x) \leq \frac{\zeta(2)}{(1-x)^2} < \frac{2}{(1-x)^2}.$$

Inserting these inequalities into (3.2) gives

$$xD(x) \geq \frac{1}{L} - 2x^2 - \frac{4x}{(1-x)^2} > 0. \quad (3.4)$$

For the last inequality, xL and x^2L increase on $(0, 1/16]$, and $\log 16 < 3$. Therefore

$$L \left(2x^2 + \frac{4x}{(1-x)^2} \right) \leq \log 16 \left(\frac{1}{128} + \frac{64}{225} \right) < \frac{25251}{28800} < 1.$$

For the remaining compact interval, set

$$p_1 = -\psi(x), \quad p_2 = -\psi(1-x), \quad v_1 = \psi'(x), \quad v_2 = \psi'(1-x), \quad f_1 = f(x), \quad f_2 = B(x).$$

The functions f_1, p_1, v_1 decrease, and f_2, p_2, v_2 increase. If $a \leq x \leq b$, interval enclosures at the two endpoints therefore give the rigorous rational lower bound

$$D(x) \geq \frac{\underline{p}_1(b)}{\underline{f}_1(a)} + \frac{\underline{p}_2(a)}{\underline{f}_2(b)} - \frac{\bar{v}_1(a)}{\underline{p}_1(b)} - \frac{\bar{v}_2(b)}{\underline{p}_2(a)}. \quad (3.5)$$

All denominators are positive. Use the 112 intervals $[j/256, (j+1)/256]$, $16 \leq j < 128$, and the explicit enclosures in Subsection 3.2. Exact rational evaluation of (3.5), reproduced by `certify_gamma_saddle.py`, proves on every interval that

$$D(x) \geq \frac{1693471917}{10^9} > 0. \quad (3.6)$$

The complete list of rational lower bounds is stored in `gamma_saddle_certificate.json`. Neither special-function floating-point evaluations nor numerical differentiation enter these checks. This finite certificate completes the inequality on $(0, 1)$.

Finally, $f(x) \sim -\log x$, $-\psi(x) \sim 1/x$, and $f(1-x) \sim \gamma x$ as $x \downarrow 0$, so $Q(x) \sim 1/\log(1/x) \rightarrow 0$. Reflection gives the limit $+\infty$ at one. The derivative of the logarithm of the product in (3.3) is $h(1-x)(1-Q(x))$. Its unique zero is $x = 1/2$, with positive derivative sign before that point and negative sign afterward. Since $\Gamma(1/2) = \sqrt{\pi}$, the sharp bound and equality case follow. $\square$

Theorem 3.2 (Uniform joint saddle expansion). *Let $\lambda = m/n$ remain in a fixed compact subset $K \subset (0, \infty)$ as $n \rightarrow \infty$. Let x_λ be the unique solution of*

$$Q(x_\lambda) = \lambda, \quad (3.7)$$

and define

$$\Phi_\lambda(x) = \log f(x) + \lambda \log f(1-x), \quad \kappa_\lambda = -\Phi_\lambda''(x_\lambda) > 0.$$

Then, uniformly for $\lambda \in K$,

$$M_{n,m} = f(x_\lambda)^n f(1-x_\lambda)^m \sqrt{\frac{2\pi}{n\kappa_\lambda}} \left(1 + \frac{C_1(\lambda)}{n} + O_K(n^{-2}) \right), \quad (3.8)$$

$$C_1(\lambda) = \frac{\Phi_\lambda^{(4)}(x_\lambda)}{8\kappa_\lambda^2} + \frac{5\Phi_\lambda^{(3)}(x_\lambda)^2}{24\kappa_\lambda^3}. \quad (3.9)$$

There is a full uniform expansion in integer powers of $1/n$. Moreover x_λ increases analytically from zero to one as λ increases from zero to infinity, and $x_{1/\lambda} = 1 - x_\lambda$.

Proof. By (3.1),

$$\Phi_\lambda'(x) = h(1-x)(\lambda - Q(x)).$$

Theorem 3.1 proves that this is positive before x_λ and negative after it. It also proves nondegeneracy:

$$\kappa_\lambda = h(1-x_\lambda)Q'(x_\lambda) > 0. \quad (3.10)$$

The analytic implicit-function theorem proves the assertions about x_λ ; reflection gives the last identity.

Compactness of K keeps x_λ a fixed distance from both endpoints, and bounds κ_λ away from zero. All required derivatives of the phase are then uniformly bounded in one fixed neighborhood of the moving saddle. The phase tends to $-\infty$ at either endpoint, uniformly for $\lambda \in K$: near zero it

is $\log \log(1/x) + \lambda \log x + O_K(1)$, and near one the analogous formula follows by reflection. Strict unimodality and compactness give a uniform positive gap between the maximum and the phase outside a fixed neighborhood of the saddle. Thus the integral over that complement is exponentially smaller than the saddle term.

In the neighborhood, put $u = \sqrt{n\kappa\lambda}(x - x_\lambda)$ and Taylor-expand the analytic phase. Gaussian domination justifies termwise integration to every fixed order. Odd powers integrate to zero; the quartic term and the square of the cubic term give (3.9). To make the error assertion explicit, expand through enough even orders on $|u| \leq n^{1/10}$, use the uniform Taylor remainder there, and bound its complement by $O(e^{-cn^{1/5}})$. This proves the stated relative error and full uniform expansion without an unproved competing-saddle assumption. $\square$

For reproducibility, the coefficient C_j at arbitrary order can be written as a finite sum. All derivatives below are evaluated at x_λ . For nonnegative integers $r_3, \dots, r_{2j+2}$ with $\sum_{k=3}^{2j+2} (k-2)r_k = 2j$, put $R = \sum_{k=3}^{2j+2} kr_k$. Then

$$C_j(\lambda) = \sum_{\sum (k-2)r_k = 2j} \frac{(R-1)!!}{\kappa_\lambda^{R/2}} \prod_{k=3}^{2j+2} \frac{1}{r_k!} \left(\frac{\Phi_\lambda^{(k)}}{k!} \right)^{r_k}, \quad (3.11)$$

with $C_0 = 1$ and $(-1)!! = 1$. The constraint makes R even. This is simply coefficient extraction after expanding the exponential of the terms of degree at least three in the local phase.

Corollary 3.3 (Equal exponents). *Set*

$$a = \frac{\log \pi}{2}, \quad p = \gamma + 2 \log 2, \quad \kappa = \frac{2}{a^2} \left(p^2 - \frac{\pi^2 a}{2} \right).$$

Then $\kappa > 0$ and

$$M_{n,n} = a^{2n} \sqrt{\frac{2\pi}{n\kappa}} \left(1 + \frac{c_*}{n} + O(n^{-2}) \right), \quad c_* = -0.4398093541949199 \dots \quad (3.12)$$

The constant is exactly $c_* = \Phi_1^{(4)}(1/2)/(8\kappa^2)$, where

$$\Phi_1^{(4)}(1/2) = 2 \left[\frac{\pi^4}{a} - \frac{56p\zeta(3) + 3\pi^4/4}{a^2} + \frac{6p^2\pi^2}{a^3} - \frac{6p^4}{a^4} \right]. \quad (3.13)$$

Numerically $\kappa = 6.2933987786891021 \dots$ and $\sqrt{2\pi/\kappa} = 0.9991882272949156 \dots$

Proof. Here $x_1 = 1/2$ and all odd phase derivatives vanish. Use $\psi(1/2) = -p$, $\psi'(1/2) = \pi^2/2$, $\psi''(1/2) = -14\zeta(3)$, and $\psi^{(3)}(1/2) = \pi^4$ in Theorem 3.2. Differentiating $\log f$ four times gives (3.13). $\square$

3.2 The compact interval certificate in full

This subsection specifies the arithmetic proof object independently of any special-function library. Every input is rational, and each infinite tail has an explicit rational bound.

For $1 \leq z \leq 2$, put $y = (z-1)/(z+1)$ and use

$$0 \leq \log z - 2 \sum_{r=0}^{31} \frac{y^{2r+1}}{2r+1} \leq \frac{2y^{65}}{65(1-y^2)}. \quad (3.14)$$

For any positive rational input, multiplication by an integral power of two reduces its logarithm to this interval; $\log 2$ is enclosed by the same formula. This controls all logarithms in the proof.

Euler's constant is enclosed with $N = 32$ by

$$H_N - \log N - \frac{1}{2N} < \gamma < H_N - \log N - \frac{1}{2N} + \frac{1}{12N^2}. \quad (3.15)$$

For a short elementary verification, set $e_N = H_N - \log N - 1/(2N)$ and $z = 1/(2N + 1)$. Expansion of $\log((1+z)/(1-z))$ gives

$$e_{N+1} - e_N = 4 \sum_{r \geq 1} \frac{r}{2r+1} z^{2r+1} > 0.$$

The sequence $u_N = e_N + 1/(12N^2)$ decreases, since

$$u_N - u_{N+1} = \sum_{r \geq 1} \frac{8r(r-1)}{3(2r+1)} z^{2r+1} > 0.$$

Both sequences tend to γ , proving (3.15). The certificate also verifies $1/2 < \gamma < 2/3$. For each integer $k \geq 2$, the elementary integral test gives

$$\sum_{r=1}^{32} \frac{1}{r^k} + \frac{33^{1-k}}{k-1} \leq \zeta(k) \leq \sum_{r=1}^{32} \frac{1}{r^k} + \frac{32^{1-k}}{k-1}. \quad (3.16)$$

In particular $1 < \zeta(k) < 2$.

For each grid endpoint $x \in [1/16, 1/2]$, use the degree-40 series

$$\begin{aligned} f_1(x) &= -\log x - \gamma x + \sum_{k=2}^{40} \frac{(-1)^k \zeta(k)}{k} x^k + \varepsilon_f, \\ f_2(x) &= \gamma x + \sum_{k=2}^{40} \frac{\zeta(k)}{k} x^k + \varepsilon_B, \\ p_1(x) &= x^{-1} + \gamma + \sum_{k=2}^{40} (-1)^{k+1} \zeta(k) x^{k-1} + \varepsilon_p, \\ p_2(x) &= \gamma + \sum_{k=2}^{40} \zeta(k) x^{k-1} + \varepsilon_s, \\ v_1(x) &= x^{-2} + \sum_{k=2}^{40} (-1)^k (k-1) \zeta(k) x^{k-2} + \varepsilon_v, \\ v_2(x) &= \sum_{k=2}^{40} (k-1) \zeta(k) x^{k-2} + \varepsilon_u. \end{aligned}$$

With

$$E_f = \frac{2x^{41}}{41(1-x)}, \quad E_p = \frac{2x^{40}}{1-x}, \quad E_v = 2x^{39} \left(\frac{40}{1-x} + \frac{x}{(1-x)^2} \right),$$

one has $|\varepsilon_f| \leq E_f$, $|\varepsilon_p| \leq E_p$, $|\varepsilon_v| \leq E_v$, and $0 \leq \varepsilon_B \leq E_f$, $0 \leq \varepsilon_s \leq E_p$, $0 \leq \varepsilon_u \leq E_v$. These follow by bounding each omitted $\zeta(k)$ by two and summing a geometric series or its derivative.

Substitute the rational intervals from (3.14)–(3.16) with the correct sign into these six formulas. Evaluate (3.5) on the 112 adjacent intervals. The smallest lower bound, rounded downward to denominator 10^9 , is the number in (3.6); it occurs on $[127/256, 1/2]$. This completely specifies the finite verification. The following command replays all the fraction operations and compares the stored proof object:

```
python certify_gamma_saddle.py -check
```

The script uses only the Python standard library and has no numerical special-function dependency. The role of the computation is the finite family of strict rational inequalities, whose analytic coverage has already been proved. Replacing it by a short conceptual proof of (3.2) is a useful further question.

3.3 The small-ratio saddle has an exponential transseries

The saddle approaches the endpoint on an exponential scale when $\lambda \downarrow 0$. Define

$$c = \frac{\zeta(2)}{2\gamma}, \quad d = c - \gamma, \quad e = \gamma^2 + \frac{\zeta(3)}{3\gamma} - \frac{\zeta(2)^2}{8\gamma^2}. \quad (3.17)$$

Here $d > 0$: $\gamma < 2/3$ and $\zeta(2) > 1$ suffice.

Proposition 3.4 (Endpoint location of the moving saddle). *As $\lambda \downarrow 0$,*

$$x_\lambda = e^{-1/\lambda} \left[1 + \left(\frac{d}{\lambda} - \gamma \right) e^{-1/\lambda} + O(\lambda^{-2} e^{-2/\lambda}) \right]. \quad (3.18)$$

The corresponding expansion near $\lambda = \infty$ follows by reflection.

Proof. For $L = \log(1/x) \rightarrow \infty$, the convergent gamma series give

$$h(x) = \frac{1}{xL} \left[1 + \gamma x + \frac{\gamma x}{L} + O(x^2) \right], \quad h(1-x) = \frac{1}{x} [1 + cx + O(x^2)].$$

Consequently

$$Q(e^{-L}) = \frac{1}{L} \left[1 - de^{-L} + \frac{\gamma e^{-L}}{L} + O(e^{-2L}) \right].$$

In the equation $Q(e^{-L}) = \lambda$, first $L \sim 1/\lambda$. Writing $L = 1/\lambda + \Delta$ and substituting once more gives

$$\Delta = \left(-\frac{d}{\lambda} + \gamma \right) e^{-1/\lambda} + O(\lambda^{-2} e^{-2/\lambda}).$$

For instance, this error follows from the mean-value theorem applied to $\lambda L - 1 + e^{-L}(d - \gamma/L) + O(e^{-2L})$, whose derivative is $\lambda + O(e^{-L})$. Exponentiating proves the proposition. $\square$

3.4 The critical window for the fixed-exponent approximation

The leading fixed- m estimate has the exact reference scale

$$L_{n,m} = \frac{\gamma^m \Gamma(n+1)}{(m+1)^{n+1}}. \quad (3.19)$$

Near the endpoint its underlying approximations are $f(x) \simeq -\log x$ and $B(x) \simeq \gamma x$. When m grows, the relative corrections to these approximations accumulate. The transition studied below is governed by $m \exp(-n/(m+1))$, not just by m/n .

Theorem 3.5 (Critical-window transition with its first correction). *Let $m \rightarrow \infty$, put*

$$t = \frac{n}{m+1}, \quad \tau = me^{-t},$$

and suppose $t - \log m$ lies in a fixed bounded interval. With c, d, e from (3.17), uniformly in that interval,

$$\frac{M_{n,m}}{L_{n,m}} = e^{d\tau} \left[1 + \frac{1}{m} \left\{ \frac{t}{2}(d\tau + d^2\tau^2) - (c + \gamma)\tau + e\tau^2 \right\} + O\left(\frac{(\log m)^2}{m^2}\right) \right]. \quad (3.20)$$

In particular, if $n/(m+1) - \log m \rightarrow s \in \mathbb{R}$, then

$$\frac{M_{n,m}}{\gamma^m \Gamma(n+1)/(m+1)^{n+1}} \rightarrow \exp(de^{-s}). \quad (3.21)$$

Thus the leading fixed- m approximation has a nonvanishing relative error throughout this window. At $s = 0$ the limiting ratio is $e^d = 2.334\dots$

Proof. Put $r = m+1$, and let T have gamma density

$$\frac{r^{n+1}}{\Gamma(n+1)} T^n e^{-rT}, \quad T > 0.$$

The substitution $x = e^{-T}$ gives the exact expectation identity

$$\frac{M_{n,m}}{L_{n,m}} = \mathbb{E}Z(T), \quad Z(T) = \left(\frac{f(e^{-T})}{T} \right)^n \left(\frac{B(e^{-T})}{\gamma e^{-T}} \right)^m. \quad (3.22)$$

Since $t - \log m$ is bounded, $t \asymp \log m$ and τ stays in a compact subinterval of $(0, \infty)$.

We first control the tails of the actual integrand. There is a constant $A > 0$ such that

$$\log \frac{B(x)}{\gamma x} \leq Ax \quad (0 < x \leq 1/2),$$

because the quotient is analytic and equals one at zero. Also $f(x)/(-\log x) \leq 1$. For $T \geq t/2$, these facts give $Z(T) \leq \exp(Ame^{-T}) \leq \exp(C\sqrt{m})$. Gamma Chernoff bounds imply, for some $c_0 > 0$,

$$\mathbb{P}(|T - t| > 1/2) \leq \exp\left(-c_0 \frac{m}{\log m}\right)$$

for sufficiently large m ; the shift between the mean and t is $1/r$. Hence the contribution with $T \geq t/2$ and $|T - t| > 1/2$ is smaller than every negative power of m . For $\log 2 \leq T < t/2$, the bound $Z(T) \leq \exp(Am/2)$ and the fixed-fraction lower-tail bound $\mathbb{P}(T < t/2) \leq \exp(-c_1 n)$ give the same conclusion, since $n \asymp m \log m$.

The remaining interval $0 < T < \log 2$ is estimated before normalization. It corresponds to $x > 1/2$. Positivity of the Taylor coefficients of B shows that $B(y)/y$ increases on $[0, 1/2]$, so

$$f(x) \leq (\log \pi)(1-x), \quad f(1-x) \leq -\log(1-x) \quad (x \geq 1/2).$$

Thus the integral over this interval is at most

$$(\log \pi)^n \int_0^{1/2} y^n (-\log y)^m dy \leq \frac{(\log \pi)^n \Gamma(m+1)}{(n+1)^{m+1}}. \quad (3.23)$$

After division by $L_{n,m}$, Stirling's formula gives logarithm $-n \log(n/m) + O(n + m \log(n/m))$. Here $n/m \asymp \log m$, so this is $-n \log \log m + O(n)$ and tends to $-\infty$ faster than any multiple of $\log m$. This controls both endpoints and justifies using a local expectation expansion on $|T - t| \leq 1/2$. The Chernoff estimates also make the omitted tails of any fixed polynomial in $T - t$ smaller than every negative power of m ; this follows either by a slightly smaller Chernoff exponent or by Cauchy–Schwarz and the gamma moments.

For that expansion write $V = T - t$ and $H(T) = \log Z(T)$. The convergent Taylor series at $x = 0$ give

$$\begin{aligned} \log \frac{B(x)}{\gamma x} &= cx + \left(\frac{\zeta(3)}{3\gamma} - \frac{c^2}{2} \right) x^2 + O(x^3), \\ \log \frac{f(x)}{-\log x} &= -\frac{\gamma x}{T} + \left(\frac{\zeta(2)}{2T} - \frac{\gamma^2}{2T^2} \right) x^2 + O(x^3/T), \quad x = e^{-T}. \end{aligned}$$

These estimates and their first four T -derivatives are uniform on $|T - t| \leq 1/2$, since there $x \asymp 1/m$ and $T \asymp \log m$. Define

$$b_2 = \frac{\zeta(3)}{3\gamma} - \frac{c^2}{2}.$$

At $T = t$ they imply

$$\begin{aligned} H(t) &= d\tau + \frac{1}{m} \left[-\gamma\tau + \tau^2 \left(b_2 + \frac{\zeta(2)}{2} - \frac{\gamma^2}{2t} \right) \right] + O(m^{-2}), \\ H'(t) &= \tau \left(-d + \frac{\gamma}{t} \right) + O(m^{-1}), \\ H''(t) &= \tau \left(d - \frac{2\gamma}{t} - \frac{2\gamma}{t^2} \right) + O(m^{-1}). \end{aligned} \tag{3.24}$$

The first four derivatives of $Z = e^H$ are bounded uniformly on the local interval. The exact gamma moments about $t = n/r$ are

$$\begin{aligned} \mathbb{E}V &= r^{-1}, \quad \mathbb{E}V^2 = t/r + 2/r^2, \\ \mathbb{E}V^3 &= 5t/r^2 + 6/r^3, \quad \mathbb{E}V^4 = 3t^2/r^2 + 26t/r^3 + 24/r^4. \end{aligned}$$

Taylor-expand $Z(t + V)$ through degree three and use the fourth absolute moment for the remainder. The signed cubic moment is $O(t/m^2)$; using an absolute cubic estimate would lose the claimed order. The tail bounds already proved permit replacement of the local moments by the complete moments. Therefore

$$\mathbb{E}Z(T) = e^{H(t)} \left[1 + \frac{H'(t)}{m} + \frac{t}{2m} (H''(t) + H'(t)^2) + O(t^2/m^2) \right]. \tag{3.25}$$

Insert (3.24). All terms containing $1/t$ cancel. The terms linear in τ reduce to $td\tau/2 - (c + \gamma)\tau$, and the quadratic terms reduce to $td^2\tau^2/2 + (b_2 + \gamma^2)\tau^2$. Since $b_2 + \gamma^2 = e$, this is (3.20). The limit follows because $t/m \rightarrow 0$ and $\tau \rightarrow e^{-s}$. $\square$

The transition condition is approximately $m \log m \asymp n$, more precisely $n = (m + 1)(\log m + s)$. The equality $m \log m = n$ has solution $m = n/W(n)$, so the nontrivial transition exhibited by the theorem lies on an $n/W(n)$ scale. This statement identifies the scale; the theorem uses the exact variables t and τ , avoiding an error from replacing $m + 1$ by m prematurely.

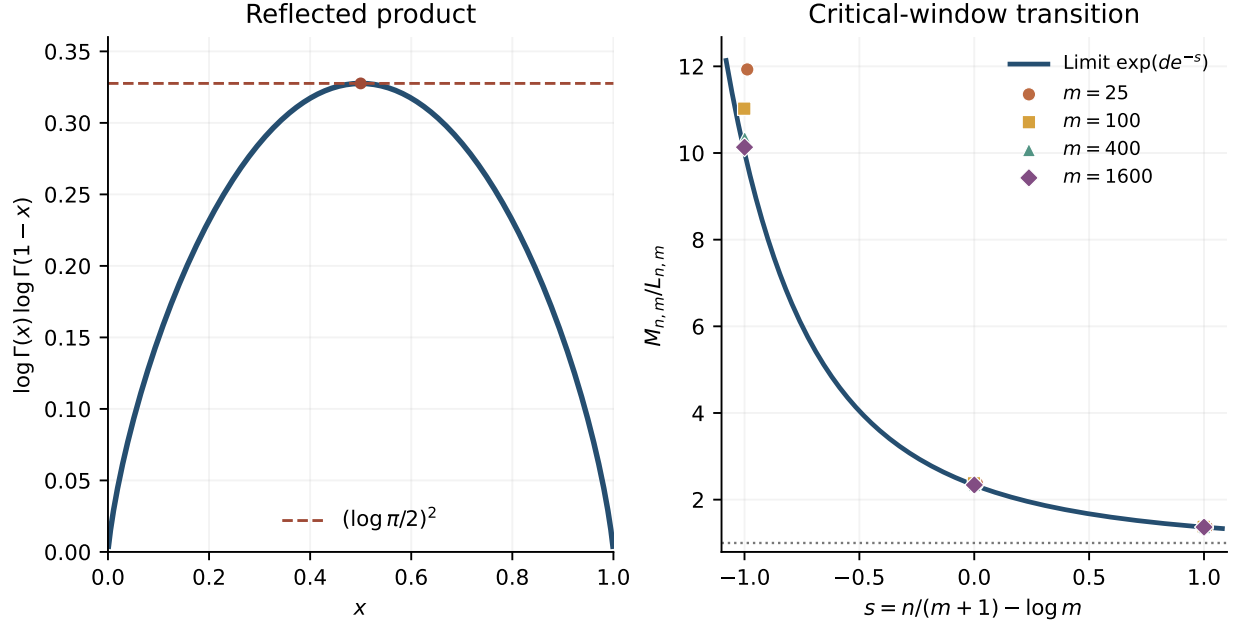


Figure 2: Left: the reflected product and its sharp maximum. Right: the critical-window profile $\exp(de^{-s})$ with numerical quadrature for $m = 25, 100, 400, 1600$. The horizontal dotted line is the fixed- m leading reference ratio one. The analytic theorem proves the profile; the plotted points are numerical diagnostics.

3.5 Numerical checks and scope

The script `check_gamma_joint_asymptotics.py` uses 65-digit quadrature and numerical differentiation solely as diagnostics. Its data are in `gamma_joint_diagnostics.json`. For equal exponents $n = 20, 40, 80$, the relative errors of the leading formula in (3.12) are approximately $2.20 \cdot 10^{-2}$, $1.10 \cdot 10^{-2}$, and $5.50 \cdot 10^{-3}$; after including c_*/n , they become $-4.76 \cdot 10^{-4}$, $-1.17 \cdot 10^{-4}$, and $-2.90 \cdot 10^{-5}$. This is consistent with the proved first and second orders.

In the critical window with $s \approx 0$, choosing $m = 100, 400, 1600$ gives $n = 465, 2403, 11812$. The relative errors of $e^{d\tau}$ are approximately $-1.67 \cdot 10^{-2}$, $-6.81 \cdot 10^{-3}$, and $-2.37 \cdot 10^{-3}$. Including the displayed correction reduces them to $-7.89 \cdot 10^{-4}$, $-1.08 \cdot 10^{-4}$, and $-1.24 \cdot 10^{-5}$, respectively. These values are not interval certificates; the analytic proof supplies the asymptotic statements.

The two proved joint regimes do not constitute one expansion uniform over all exponent ratios. Section 7 specifies the remaining matching, coefficient, and optimal-truncation questions.

4 Lerch deformation: loss of real zeros and forced bifurcations

The manuscript proves eventual saturation of the positive-zero bound as the parameter-derivative order grows, uniformly in the Lerch deformation [1]. At a fixed small derivative order the geometry is different. We prove an exact count for sufficiently small positive deformation parameters. Multiple elementary roots generally leave the real axis. At indices three and four this forces an interior change in the number of positive zeros before the Stieltjes endpoint is reached.

For integers $n \geq 0$, $k \geq 1$, let

$$f_{n,k}(a) = \frac{d^k}{da^k} \frac{\log^n a}{a}, \quad D_{n,k}^\rho(a) = \sum_{m=0}^{\infty} \rho^m f_{n,k}(a+m) \quad (a > 0, 0 \leq \rho \leq 1). \quad (4.1)$$

Here $0^0 = 1$ in the $m = 0$ summand. The differentiated series is absolutely convergent even at $\rho = 1$, and $D_{n,k}^1(a) = \gamma_n^{(k)}(a)$ for the Laurent convention

$$\zeta(s, a) = \frac{1}{s-1} + \sum_{n \geq 0} \frac{(-1)^n}{n!} \gamma_n(a) (s-1)^n.$$

For $\rho < 1$, it equals $\partial_a^k \ell_n(\rho, a)$ with $\ell_n(\rho, a) = \sum_{m \geq 0} \rho^m \log^n(a+m)/(a+m)$. In particular, these are order derivatives of the Lerch transcendent; at $a = 1$ their undeformed-in- a antecedents specialize to order derivatives of $\text{Li}_s(\rho)$.

Define the rational polynomial

$$P_{n,k}(x) = n! [t^n] \left(e^{xt} \prod_{j=1}^k (1+t/j) \right). \quad (4.2)$$

Coefficient extraction after differentiating a^{-1-t} gives

$$(-1)^{n+k} f_{n,k}(a) = k! a^{-k-1} P_{n,k}(-\log a). \quad (4.3)$$

The sign normalization $F_{n,k}^\rho = (-1)^{n+k} D_{n,k}^\rho$ does not affect zeros.

4.1 The elementary roots and uniform exclusion of escaping zeros

Lemma 4.1. *If $n > k \geq 1$ and $d = n - k$, then $P_{n,k}$ has a root of multiplicity d at zero and precisely k other roots, all simple and negative. Its first nonzero coefficient is*

$$[x^d] P_{n,k}(x) = \frac{n!}{d! k!}.$$

Proof. The polynomial is $\prod_{j=1}^k (1 + \partial_x/j) x^n$. If a polynomial p has real roots r_i of multiplicities m_i , the new noninherited roots of $p + cp'$ solve

$$\frac{p'(x)}{p(x)} = \sum_i \frac{m_i}{x - r_i} = -1/c.$$

For $c > 0$ the left side decreases strictly between its poles. There is one new simple root in every intervening interval and one to the left of all the old roots, and there is none to their right. Every inherited multiple root loses one unit of multiplicity. Induction from x^n therefore gives the asserted configuration. The coefficient follows from the top coefficient $1/k!$ in $\prod_{j=1}^k (1 + t/j)$. $\square$

Let $A_{n,k} > 1$ be the largest zero of $f_{n,k}$, equivalently the exponential of the negative of the smallest root of $P_{n,k}$.

Lemma 4.2 (No escape through either endpoint). *For fixed $n > k \geq 1$ there is $\varepsilon_{n,k} > 0$ such that every positive zero of $D_{n,k}^\rho$, for every $\rho \in [0, 1]$, lies in $[\varepsilon_{n,k}, A_{n,k}]$. For $\rho > 0$ the upper inequality is strict. Moreover $D_{n,k}^\rho \rightarrow f_{n,k}$ uniformly on every compact subinterval of $(0, \infty)$ as $\rho \downarrow 0$, including every fixed number of a -derivatives.*

Proof. The $m = 0$ term in the normalized series satisfies

$$F_{n,k}^0(a) \sim k!a^{-k-1} \log^n(1/a) \longrightarrow +\infty \quad (a \downarrow 0).$$

For $0 < a \leq 1$, the sum of the absolute values of all $m \geq 1$ terms is bounded independently of $\rho \in [0, 1]$: each is bounded by a constant times $(1 + \log^n(m+1))/m^{k+1}$, whose series converges. Thus the $m = 0$ term dominates uniformly near zero.

For $a > A_{n,k}$, the argument $-\log(a+m)$ lies strictly to the left of every root of $P_{n,k}$ for all $m \geq 0$. Consequently all summands have the same nonzero sign. At $a = A_{n,k}$ the $m = 0$ summand vanishes and the others have that common strict sign when $\rho > 0$. This excludes the upper endpoint and infinity. On a compact positive a -interval, the same summable majorants, with k increased for the derivatives, give the final assertion by dominated convergence. $\square$

4.2 Complete small-parameter count and Puiseux branches

Theorem 4.3 (Unfolding the multiple elementary root). *Fix $n > k \geq 1$, put $d = n - k$, and define*

$$A = \frac{n!}{d!}, \quad B = \frac{k!}{2^{k+1}} P_{n,k}(-\log 2). \quad (4.4)$$

Assume $B \neq 0$. For all sufficiently small $\rho > 0$, every positive zero of $D_{n,k}^\rho$ is simple and their exact number is

$$k + \begin{cases} 1, & d \text{ odd}, \\ 2, & d \text{ even and } B < 0, \\ 0, & d \text{ even and } B > 0. \end{cases} \quad (4.5)$$

The k zeros continuing the elementary zeros different from one are real analytic functions of ρ . All d complex zeros near $a = 1$ are convergent Puiseux branches. Writing $t = \rho^{1/d}$ and letting $y_1, \dots, y_d$ be the distinct roots of

$$Ay^d + B = 0, \quad (4.6)$$

these branches have the form

$$a_j(\rho) = 1 - y_j \rho^{1/d} + O(\rho^{2/d}). \quad (4.7)$$

Their series converge for sufficiently small t . In particular, if $d \geq 3$, the number in (4.5) is strictly less than n .

Proof. For $|\rho| < 1$ the series is normally convergent on compact subsets of $\Re a > 0$ and hence jointly holomorphic in (a, ρ) . The principal logarithm is unambiguous on this half-plane. The implicit-function theorem continues each of the k nonunit elementary zeros uniquely and real analytically.

Near the remaining zero use the analytic coordinate $a = e^{-x}$. Equations (4.3) and (4.4) give

$$F_{n,k}^\rho(e^{-x}) = Ax^d + B\rho + O(x^{d+1} + \rho x + \rho^2). \quad (4.8)$$

Substitute $\rho = t^d$ and $x = ty$, divide by t^d , and extend analytically to $t = 0$. The resulting function equals $Ay^d + B$ at zero. Its d roots are simple because $AB \neq 0$, so the implicit-function theorem supplies d analytic branches $y_j(t) = y_j + O(t)$. Rouché's theorem on a fixed small circle around $x = 0$ shows that these exhaust the zeros converging to zero. A branch is real for small positive t exactly when its leading root y_j is real: real leading roots continue by conjugation and uniqueness;

nonreal leading roots stay nonreal. The polynomial $Ay^d + B$ has precisely the real-root count displayed in (4.5), since $A > 0$.

Finally, Lemma 4.2 confines every positive zero to a fixed compact interval. Delete disjoint small neighborhoods of the elementary zeros from that interval. The limit $f_{n,k}$ is bounded away from zero on the remaining compact set; uniform convergence excludes every additional zero there. Thus the local branch count is the global positive-zero count. Simplicity follows from the nonzero implicit derivatives, at both the ordinary and rescaled branches. $\square$

The nonvanishing condition is displayed explicitly in the general statement. For the following first-derivative specialization, its sign and nonvanishing have an elementary proof for every index.

Corollary 4.4 (All indices at first derivative order). *For every fixed integer $n \geq 2$, and all sufficiently small $\rho > 0$, $D_{n,1}^\rho$ has exactly one positive zero if n is odd and exactly two if n is even. All are simple. One zero tends to e^n and depends real analytically on ρ . When n is even, the second zero satisfies*

$$a_{\text{small}}(\rho) = 1 - \kappa_n \rho^{1/(n-1)} + O(\rho^{2/(n-1)}), \quad \kappa_n = \log 2 \left(\frac{n - \log 2}{4n} \right)^{1/(n-1)} > 0. \quad (4.9)$$

The root tending to e^n moves initially to the left:

$$\begin{aligned} a_{\text{large}}(\rho) &= e^n - C_n \rho + O(\rho^2), \\ C_n &= \frac{e^{3n} \log^{n-1}(e^n + 1)(\log(e^n + 1) - n)}{n^{n-1}(e^n + 1)^2} > 0. \end{aligned} \quad (4.10)$$

Proof. Here

$$P_{n,1}(x) = x^{n-1}(x + n), \quad f_{n,1}(a) = \frac{\log^{n-1} a (n - \log a)}{a^2}.$$

Since $0 < \log 2 < 1 < n$, the constant B has sign $(-1)^{n-1}$. For odd n the multiplicity $n - 1$ is even and $B > 0$, giving no real branch near one. For even n it is odd and $B < 0$, giving one positive leading root $y = \kappa_n$ in (4.6). This proves the count and (4.9). At $a = e^n$, $f'_{n,1}(e^n) = -n^{n-1}e^{-3n}$. Implicit differentiation of $f_{n,1}(a) + \rho f_{n,1}(a + 1) + O(\rho^2) = 0$ gives $a'(0) = -f_{n,1}(e^n + 1)/f'_{n,1}(e^n) = -C_n$. $\square$

For example, the double elementary zero for $n = 3$ gives a conjugate nonreal pair with $a = 1 \mp i\kappa_3\sqrt{\rho} + O(\rho)$, whereas for $n = 4$ the triple zero gives one real branch below one and a nonreal conjugate pair. The manuscript's eventual simplicity theorem is fully consistent with this result: its derivative order tends to infinity, while the present theorem fixes the derivative order and lets the deformation tend to zero.

4.3 The logarithmic singularity at the Stieltjes endpoint

An elementary tail calculation supplies a complementary expansion as $\rho \uparrow 1$. It gives the precise regularity that is not visible in continuity of the positive-kernel representation. In this subsection the indices are fixed and the limit concerns the deformation parameter.

For $0 \leq j < k$ define the absolutely convergent moments

$$M_{n,k,j}(a) = \sum_{m=0}^{\infty} m^j f_{n,k}(a + m), \quad M_{n,k,0}(a) = \gamma_n^{(k)}(a). \quad (4.11)$$

The $m = 0$ factor is one when $j = 0$ and zero otherwise.

Theorem 4.5 (Endpoint expansion with its first nonanalytic term). *Fix integers $n \geq 0$, $k \geq 1$ and a compact interval $I \subset (0, \infty)$. Put $\epsilon = -\log \rho$ and $L = \log(1/\epsilon)$. As $\epsilon \downarrow 0$, uniformly for $a \in I$,*

$$D_{n,k}^{\epsilon^{-\epsilon}}(a) = \sum_{j=0}^{k-1} \frac{(-\epsilon)^j}{j!} M_{n,k,j}(a) + \frac{\epsilon^k L^{n+1}}{n+1} + O_{n,k,I}(\epsilon^k(1+L^n)). \quad (4.12)$$

Thus the family is C^{k-1} at the endpoint in the deformation parameter, but its k th one-sided derivative is not finite there. In particular it is not C^k . The coefficient of the leading nonanalytic term is independent of a and k .

Proof. For large m , uniformly on I , differentiation of $\log^n(a+m)/(a+m)$ gives

$$f_{n,k}(a+m) = (-1)^k k! \frac{\log^n m}{m^{k+1}} + E_{n,k}(a, m). \quad (4.13)$$

For $n \geq 1$ the error is bounded by a constant times

$$\frac{1 + \log^{n-1} m}{m^{k+1}} + \frac{1 + \log^n m}{m^{k+2}};$$

for $n = 0$ it is $O(m^{-k-2})$. Indeed, the leading term arises when all k derivatives fall on $1/(a+m)$; every differentiation of a logarithm lowers its degree, and replacement of $a+m$ by m adds one inverse power. This also proves convergence and local uniform convergence of all the moments in (4.11).

Write

$$r_k(v) = e^{-v} - \sum_{j=0}^{k-1} \frac{(-v)^j}{j!}.$$

Taylor's theorem and the finite polynomial bound give

$$\begin{aligned} r_k(v) &= \frac{(-1)^k v^k}{k!} + O_k(v^{k+1}) & (0 \leq v \leq 1), \\ |r_k(v)| &\leq C_k v^{k-1} & (v \geq 1). \end{aligned} \quad (4.14)$$

Subtract the finite moment sum in (4.12) from the defining series. The remainder is exactly $\sum_{m \geq 1} f_{n,k}(a+m)r_k(\epsilon m)$. Put $N = \lfloor 1/\epsilon \rfloor$. In the range $m \leq N$ the error from the first line of (4.14) is bounded by

$$C\epsilon^{k+1} \sum_{m \leq N} m^{k+1} |f_{n,k}(a+m)| \leq C'\epsilon^k(1+L^n).$$

Its retained part is

$$\frac{(-1)^k \epsilon^k}{k!} \sum_{m \leq N} m^k f_{n,k}(a+m).$$

By (4.13), ordinary integral comparison for logarithmic harmonic sums gives

$$\sum_{m \leq N} m^k f_{n,k}(a+m) = (-1)^k k! \left\{ \frac{L^{n+1}}{n+1} + O(1+L^n) \right\}.$$

For $n = 0$ the error contributed by (4.13) is bounded; for $n \geq 1$ its lower logarithmic terms sum to $O(1 + L^n)$. The replacement of $\log N$ by L is absorbed in the same error. The two factors $(-1)^k$ cancel, producing the positive coefficient in the theorem.

Finally the range $m > N$, using the second bound in (4.14), is at most

$$C\epsilon^{k-1} \sum_{m>N} \frac{1 + \log^n m}{m^2} = O(\epsilon^k(1 + L^n)).$$

This proves the expansion with the claimed uniformity.

Differentiation with respect to ϵ through order $k-1$ is justified at zero by the absolutely convergent weighted moment series, uniformly on I . For clarity, the failure of the next derivative can also be checked directly, without differentiating a big- O term. Put $G(\epsilon) = D_{n,k}^{e^{-\epsilon}}(a)$. Then

$$G^{(k-1)}(\epsilon) = (-1)^{k-1} \sum_{m \geq 0} m^{k-1} e^{-\epsilon m} f_{n,k}(a+m).$$

Apply the same split at $m = \lfloor 1/\epsilon \rfloor$ to the difference between this expression and its value at zero. Its summand has leading coefficient $-k! \log^n m / m^2$, by (4.13), and therefore

$$\frac{G^{(k-1)}(\epsilon) - G^{(k-1)}(0)}{\epsilon} = \frac{k!}{n+1} L^{n+1} + O(1 + L^n) \rightarrow +\infty.$$

This proves the failure of a finite k th one-sided derivative. The change of coordinates $\rho = e^{-\epsilon}$ is smooth with nonzero derivative at zero; its lower-order chain-rule terms have finite limits. Thus the same endpoint regularity holds in ρ . $\square$

Corollary 4.6 (Displacement of a simple Stieltjes zero). *Fix an integer $n \geq 1$. Let $a_* > 0$ be a simple zero of $\gamma'_n(a)$, so $\gamma''_n(a_*) \neq 0$. For $\rho < 1$ sufficiently close to one there is a unique nearby simple zero $a_*(\rho)$ of $D_{n,1}^\rho$. With $\epsilon = -\log \rho$ and $L = \log(1/\epsilon)$,*

$$a_*(\rho) = a_* - \frac{\epsilon L^{n+1}}{(n+1)\gamma''_n(a_*)} + O(\epsilon(1 + L^n)). \quad (4.15)$$

Proof. Local uniform convergence of the series and its a -derivative gives a unique simple zero near a_* . The $k = 1$ specialization of Theorem 4.5 is

$$D_{n,1}^{e^{-\epsilon}}(a) - \gamma'_n(a) = \frac{\epsilon L^{n+1}}{n+1} + O(\epsilon(1 + L^n)),$$

uniformly on a small interval around a_* . A derivative bounded away from zero first gives $a_*(\rho) - a_* = O(\epsilon L^{n+1})$. Taylor expansion of γ'_n at a_* then gives the displayed formula. Its quadratic error is absorbed because $\epsilon^2 L^{2n+2} = o(\epsilon(1 + L^n))$. $\square$

The logarithmic enhancement means that a very small value of $1 - \rho$ can still noticeably displace a fixed-index zero. This is a proved endpoint scale. It does not determine where an interior multiple root occurs, because the simple-root expansion ceases to apply when the derivative in its denominator approaches zero.

4.4 Fresh exact endpoint signs

We recall, with proof, the global zero bound used to upgrade sign changes to exact endpoint counts. It is the result entitled “Global bound on positive zeros” in Chapter 10 of the source manuscript; the accompanying results are entitled “Strictly real-rooted reciprocal-gamma polynomials” and “Appell–Laplace representation” [1]. Define reciprocal-gamma Appell polynomials Q_n by

$$\frac{e^{xt}}{\Gamma(1+t)} = \sum_{n \geq 0} Q_n(x) \frac{t^n}{n!}.$$

Lemma 4.7 (Global positive-zero bound). *For every $n \geq 0$, $k \geq 1$, and $0 \leq \rho \leq 1$, the function $D_{n,k}^\rho$ has at most n positive zeros, counted with multiplicity.*

Proof. The Weierstrass product is

$$\frac{1}{\Gamma(1+t)} = e^{\gamma t} \prod_{j \geq 1} (1 + t/j) e^{-t/j}.$$

Taking $n![t^n]$ after multiplication by e^{xt} turns finite products into the operators $1 + \partial_x/j$ and real translations applied to x^n . These preserve real-rootedness by the operator proof in Lemma 4.1. Local uniform convergence of the product gives coefficient convergence of the resulting monic polynomials, so the limiting Q_n is real-rooted. For $n \geq 1$, remove the first $n-1$ factors before taking the limit. The remaining tail still gives a real-rooted monic polynomial. Restoring the removed factors, including their commuting real translations, applies $n-1$ root-preserving operators, which reduce every inherited multiplicity until all roots are simple. Thus Q_n has exactly n distinct real roots. For $n=0$ one has $Q_0=1$.

The Mellin integral gives the identity

$$\sum_{n \geq 0} \frac{(-1)^{n+k}}{n!} D_{n,k}^\rho(a) t^n = \frac{1}{\Gamma(1+t)} \int_0^\infty \frac{x^{k+t} e^{-ax}}{1 - \rho e^{-x}} dx$$

for $|t| < k$ in a neighborhood of zero. It follows either by differentiating $(a+m)^{-1-t}$ k times, or by canceling $\Gamma(k+1+t)$ against its rising factorial. Coefficient comparison proves

$$F_{n,k}^\rho(a) = \int_0^\infty \frac{x^k e^{-ax}}{1 - \rho e^{-x}} Q_n(\log x) dx. \quad (4.16)$$

At zero, $1/(1 - \rho e^{-x}) \leq 1/(1 - e^{-x}) = O(x^{-1})$, so $x^{k-1} |\log x|^n$ supplies an integrable majorant. At infinity e^{-ax} dominates all the logarithmic and polynomial factors. These estimates justify coefficient extraction, differentiation in a , and multiplication of the kernel by any fixed polynomial.

The signed kernel in (4.16) has exactly n sign changes. For completeness, let G be a Laplace transform of such a kernel. If there are no sign changes, G has a fixed strict sign. Otherwise choose one sign-change point x_0 . Multiplication of the kernel by $x_0 - x$ removes that sign change and introduces none. Its transform is $(\partial_a + x_0)G = e^{-x_0 a} (e^{x_0 a} G)'$ and, by induction, has at most $n-1$ zeros counted with multiplicity and is not identically zero. Therefore G also cannot be identically zero. Any finite collection of M zeros of G , counted with multiplicity, forces at least $M-1$ zeros of $(e^{x_0 a} G)'$ by Rolle’s theorem, including inherited multiplicities. Hence $M \leq n$. The assertion holds for every finite collection and therefore for all positive zeros, completing the induction and proof. $\square$

The following are newly replayed rational enclosures for

$$V_n(a) = (-1)^{n+1} a^2 \gamma'_n(a).$$

Each displayed decimal endpoint is an exact terminating rational number.

n	a	Interval containing $V_n(a)$	Sign
3	4/5	[0.163799, 0.163919]	+
3	11/10	[-0.041066, -0.040839]	-
3	3/2	[0.055828, 0.056247]	+
3	21/10	[-0.245836, -0.245015]	-
4	4/5	[0.038668, 0.039282]	+
4	47/50	[-0.003812, -0.002965]	-
4	6/5	[0.021074, 0.022453]	+
4	17/10	[-0.037951, -0.035186]	-
4	11/5	[0.148603, 0.153231]	+

For reproducibility, the included script uses only Python's standard library and outward integer arithmetic on the grid $2^{-160}\mathbb{Z}$. It sums the first $M = 1000$ terms of (4.1) at $k = 1, \rho = 1$. For a positive rational argument, power-of-two range reduction brings the logarithm to $1 \leq x \leq 2$. With $u = (x - 1)/(x + 1) \in [0, 1/3]$, it uses

$$0 \leq \log x - 2 \sum_{j=0}^{J-1} \frac{u^{2j+1}}{2j+1} \leq \frac{2u^{2J+1}}{(2J+1)(1-u^2)}, \quad J = 64. \quad (4.17)$$

All powers, divisions and the geometric remainder are enclosed by integer floor and ceiling. Inverting the argument handles logarithms below one. Ordinary interval multiplication then evaluates $f_{n,1}$.

The tail needs only a monotone integral comparison. Put $b = a + M$ and $h_n(x) = -f_{n,1}(x)$. For $n = 3, 4$ and $\log x \geq 6$, this function is positive and decreasing, since

$$h'_n(x) = -\frac{\log^{n-2} x}{x^3} \{2 \log^2 x - 3n \log x + n(n-1)\} < 0.$$

Every argument b above satisfies $\log b > 6$. As $\int_b^\infty h_n(x) dx = \log^n b / b$, one has the exact enclosure

$$f_{n,1}(b) - \frac{\log^n b}{b} \leq \sum_{m=M}^{\infty} f_{n,1}(a+m) \leq -\frac{\log^n b}{b}. \quad (4.18)$$

Combining these bounds proves every displayed sign; the JSON artifact also contains the finer integer endpoints before decimal display rounding. Thus these signs do not depend on quadrature or a special-function library.

Corollary 4.8 (Forced interior multiple roots). *For each $n \in \{3, 4\}$ there are $\rho_* \in (0, 1)$ and $a_* > 0$ such that*

$$D_{n,1}^{\rho_*}(a_*) = 0, \quad \partial_a D_{n,1}^{\rho_*}(a_*) = 0. \quad (4.19)$$

At $\rho = 1$ there are exactly n positive zeros, all simple, and this count persists for $\rho < 1$ sufficiently close to one. For small positive ρ the counts are instead one and two, respectively.

Proof. The certified signs give n disjoint intervals with sign changes. The global bound with multiplicity shows that these are all the roots and each is simple. Local uniform convergence as $\rho \uparrow 1$ of the series and its a -derivative preserves a unique simple root in each small interval. The global bound excludes extras. The small-parameter counts are Corollary 4.4.

If every root were simple throughout $0 < \rho < 1$, the implicit-function theorem and compact exclusion of both endpoints would make the number of positive roots locally constant on this connected interval, hence constant. More explicitly, finitely many small root neighborhoods and the zero-free compact complement yield a common parameter neighborhood on which no root can be added or lost. The different endpoint counts contradict constancy. Therefore (4.19) holds at an interior parameter. $\square$

Ordinary numerical quadrature of (4.16), followed by a two-variable solve, suggests transitions near

n	a_*	ρ_*
3	1.0497530509	0.99999046684
4	1.2013268276	0.99999992971

These coordinates are numerical observations. The theorem establishes an interior multiple root; it does not establish uniqueness of the transition, multiplicity exactly two, or a nonzero derivative in the deformation parameter. Certifying those stronger statements is a specific further problem, approachable by a two-variable interval Newton calculation followed by an analytic exclusion argument.

5 A weight-nine candidate and exact proximity certificates

The source now proves its mixed Gaussian weight-five identity for S_4 by an octahedral change of variables and an exact rational certificate. Its corresponding reduction for S_6 remains a conjecture [1]. We find the next even-index candidate, S_8 , in the same deliberately small family of single-color double values and products. The new identity in this section is conjectural. The rational error enclosures below are proved statements about its proximity to equality.

5.1 The candidate and its frozen integer vector

We use the source conventions

$$S_p = \sum_{n \geq 0} \frac{(-1)^n H_n}{(2n+1)^p}, \quad g_{a,b} = \operatorname{Im} \operatorname{Li}_{a,b}(i, 1) = \sum_{n \geq 0} \frac{(-1)^n H_{2n}^{(b)}}{(2n+1)^a}, \quad G = \beta(2),$$

where $H_n^{(b)} = \sum_{j=1}^n j^{-b}$ and $\beta(s) = \sum_{n \geq 0} (-1)^n / (2n+1)^s$.

Conjecture 5.1 (A weight-nine mixed Gaussian reduction). *The following identity holds:*

$$\begin{aligned} S_8 = & \frac{52334}{37973} g_{8,1} + \frac{2824}{37973} g_{6,3} - \frac{6832}{189865} g_{4,5} - \frac{139760}{265811} g_{2,7} \\ & + \frac{998183}{17283022848} \pi^9 - \frac{78615}{19442176} G \zeta(7) - \frac{201021}{4252976} \beta(4) \zeta(5) \\ & - \frac{290505}{1063244} \beta(6) \zeta(3) - 2\beta(8) \log 2. \end{aligned} \tag{5.1}$$

In the ordered basket

$$(S_8, g_{8,1}, g_{6,3}, g_{4,5}, g_{2,7}, \pi^9, G\zeta(7), \beta(4)\zeta(5), \beta(6)\zeta(3), \beta(8)\log 2), \quad (5.2)$$

the frozen primitive integer vector is

$$\begin{aligned} q = (& -10974719508480, 15125246115840, 816174858240, \\ & -394908401664, -5770366156800, 633846205, \\ & -44376595200, -518730670080, -2998569369600, \\ & -21949439016960). \end{aligned} \quad (5.3)$$

Its entries have greatest common divisor one. All coefficients in (5.1) are $-q_j/q_0$; thus their signs and denominators can be checked independently of the displayed fractions. No independence or minimality claim is made for the basket.

5.2 Discovery and independent numerical audits

The discovery evaluation used 720 Euler terms at 250 working decimal digits. PSLQ was then run at 150 working digits, with tolerance 10^{-140} , coefficient bound 10^{20} , and a maximum of 10000 steps. The largest entry of the vector actually returned is less than 2.2×10^{13} . After freezing the vector, a fresh 1400-term Euler evaluation at 500 working digits gave the integer-normalized residual

$$q \cdot \mathbf{v} \simeq -1.65201554251472773306398896477 \times 10^{-409}. \quad (5.4)$$

The truncation length, rather than the 500-digit arithmetic, is material here. This diagnostic is not asserted to bound the true residual.

For a separate computational route, the Mellin integrals

$$S_8 = -\frac{1}{7!} \int_0^1 \frac{(-\log x)^7 \log(1+x^2)}{1+x^2} dx, \quad (5.5)$$

$$g_{a,b} = \frac{1}{(a-1)!} \int_0^1 (-\log x)^{a-1} \operatorname{Im} \frac{i \operatorname{Li}_b(ix)}{1-ix} dx \quad (5.6)$$

were evaluated directly at 220 working digits, using $x = e^{-t}$. They do not use the Euler weights or the finite harmonic sums of the discovery computation. The resulting integer-normalized residual was

$$7.75350546594884491938636345482 \times 10^{-208},$$

or $-7.06487802258 \times 10^{-221}$ after division by q_0 . Both evaluations give

$$S_8 = -0.00014885380029854835375941306724994026465756408905187697 \dots$$

The integral was truncated at $t = 256$. A direct geometric bound gives, for the Gaussian integral of outer exponent a , the discarded-tail bound

$$\frac{2\Gamma(a, 768)}{(a-1)! 3^a (1 - e^{-512})}.$$

For the four exponents used here this is less than 3×10^{-321} ; the same largest bound also controls the S_8 tail. The quadrature on the finite interval is ordinary high-precision numerical quadrature, so the displayed residual is an audit, not an interval certificate.

5.3 A short self-contained Euler bound

The following elementary bound avoids the signed-density analysis of the source. It is somewhat weaker, but particularly convenient for a standalone exact computation. For a sequence f define

$$(Df)(n) = f(n) - f(n+1), \quad E_N(f) = \sum_{k=0}^{N-1} \frac{D^k f(0)}{2^{k+1}}.$$

Proposition 5.2 (Direct one-sided Euler enclosures). *For integers $a, p \geq 2$, $b \geq 1$, and $N \geq 1$, put $f_{a,b}(n) = H_{2n}^{(b)}/(2n+1)^a$ and $f_p^S(n) = H_n/(2n+1)^p$. Then*

$$E_N(f_{a,b}) - \frac{(N+1)(1+2^{-b})}{3^a 2^N} \leq g_{a,b} \leq E_N(f_{a,b}), \quad (5.7)$$

$$E_N(f_p^S) - \frac{N+1}{3^p 2^N} \leq S_p \leq E_N(f_p^S). \quad (5.8)$$

All finite differences of positive order occurring here are strictly negative. The finite approximants decrease to their respective limits.

Proof. The elementary moment formulas give

$$D^k f_{a,b}(0) = \frac{1}{\Gamma(a)\Gamma(b)} \int_0^1 \int_0^1 (-\log u)^{a-1} (-\log v)^{b-1} \times \frac{(1-u^2)^k - (1-u^2 v^2)^k}{1-v} dv du.$$

The numerator is negative for $k \geq 1$ and $0 < u, v < 1$. By the mean-value theorem its absolute value is at most $ku^2(1-v^2)$. Consequently

$$|D^k f_{a,b}(0)| \leq \frac{k}{3^a} \left(1 + \frac{1}{2^b}\right).$$

For f_p^S , the representation $H_n = \int_0^1 (1-v^n)/(1-v) dv$ gives the same double integral with v^2 replaced by v and no logarithmic v factor. Thus $|D^k f_p^S(0)| \leq k/3^p$, again with a negative sign.

These bounds justify absolute summation of the Euler kernels under the integrals. Summing the geometric kernels identifies the limits with the defining alternating sums; one may equivalently begin with a radial factor and pass to its limit. Finally

$$\sum_{k=N}^{\infty} \frac{k}{2^{k+1}} = (N+1)2^{-N},$$

which proves the two one-sided error bounds and the strict decrease. $\square$

The finite rearrangement

$$E_N(f) = 2^{-N} \sum_{n=0}^{N-1} (-1)^n f(n) \sum_{j=n+1}^N \binom{N}{j} \quad (5.9)$$

follows by expanding $D^k f(0)$ and using

$$\sum_{k=n}^{N-1} 2^{N-k-1} \binom{k}{n} = \sum_{j=n+1}^N \binom{N}{j}.$$

This identity can be proved directly by Pascal's recurrence. It avoids both numerical finite differences and a quadratic-size rational table. For example, if $L_N = \text{lcm}(1, \dots, 2N)$, all summands for $g_{a,b}$ have the common denominator $2^N L_N^{a+b}$. The program forms integer harmonic numerators, integer binomial tails, and one final exact rational quotient for each Euler approximant.

5.4 Exact rational proximity for both candidates

Let Δ_8 denote the left side of (5.1) minus its right side. For completeness, let Δ_6 be the same difference for the source candidate

$$S_6 \stackrel{?}{=} \frac{722}{527}g_{6,1} + \frac{40}{527}g_{4,3} - \frac{128}{155}g_{2,5} + \frac{15191}{28569600}\pi^7 - \frac{3}{124}G\zeta(5) - \frac{2373}{10540}\beta(4)\zeta(3) - 2\beta(6)\log 2. \quad (5.10)$$

Proposition 5.3 (Certified proximity, not equality). *The defining constants satisfy*

$$-10^{-355} \leq \Delta_6 \leq 10^{-355}, \quad -10^{-355} \leq \Delta_8 \leq 10^{-355}. \quad (5.11)$$

These bounds do not prove either candidate identity.

Proof. Use $N = 1200$ in Proposition 5.2 and evaluate (5.9) in exact integer arithmetic. The remaining constants admit independent elementary rational enclosures. The sequences $(2n+1)^{-s}$ and $(n+1)^{-s}$ are moments of positive densities of mass one. Their D -differences belong to $[0, 1]$, so their N -term Euler sums enclose $\beta(s)$ and $\eta(s)$ from below, with error at most 2^{-N} . Dividing the second interval by $1 - 2^{1-s}$ encloses $\zeta(s)$ for $s > 1$.

For π , use Machin's identity $\pi = 16 \arctan(1/5) - 4 \arctan(1/239)$ and 300 terms of each alternating arctangent series, whose first omitted term bounds its remainder. The identity itself follows by tangent addition and the location of the angle in $(0, \pi/2)$. For $\log 2$, use

$$0 < \log 2 - 2 \sum_{n=0}^{399} \frac{1}{(2n+1)3^{2n+1}} \leq \frac{9}{4 \cdot 801 \cdot 3^{801}}.$$

The general bound follows by replacing each denominator in the tail of $2 \operatorname{arctanh}(1/3)$ by the first omitted denominator and summing the geometric series.

Rational interval addition and multiplication now enclose the two differences. The accompanying `interval_certificate.py` recomputes every endpoint from these defining formulas using Python integers and `Fraction`; it uses no floating-point input, PSLQ, cached polylogarithms, or quadrature values. Exact cross multiplication checks both inclusions in (5.11). The complete rational endpoints and outward decimal displays are recorded in `s6_s8_direct_interval_certificate.json`. $\square$

A second computation at $N = 1100$ uses the sharper source estimate

$$E_N(f_{a,b}) - C_b 2^{-N} \leq g_{a,b} \leq E_N(f_{a,b}), \quad C_1 = 1, \quad C_b = b/(b-1) \ (b \geq 2),$$

whose proof uses the signed kernel of [1]. It independently checks the same exact-arithmetic assembly and gives $\Delta_6, \Delta_8 \in [-10^{-330}, 10^{-330}]$. The exact intervals in that computation are approximately

$$\Delta_6 \in [-2.44503, 1.99357] \times 10^{-331}, \quad \Delta_8 \in [-1.49934, 1.81452] \times 10^{-331}.$$

The word “approximately” applies only to these displayed endpoint roundings; the accompanying JSON stores the full exact fractions. Interval overlap is consistent with equality and detects many errors, but it cannot exclude a sufficiently small nonzero difference.

5.5 The next structural question

There is a common pattern in the proved S_4 identity, the source's S_6 conjecture, and the new S_8 conjecture. The source's proved weight-five shuffle relations rewrite its S_4 theorem as

$$S_4 = \frac{10}{7}g_{4,1} - \frac{8}{7}g_{2,3} + \frac{7}{1536}\pi^5 - \frac{3}{28}G\zeta(3) - 2\beta(4)\log 2. \quad (5.12)$$

For $m \geq 2$, define the finite family

$$\begin{aligned} \mathcal{B}_m = & \{g_{2m-2j,2j+1} : 0 \leq j < m\} \cup \{\pi^{2m+1}\} \\ & \cup \{\beta(2j)\zeta(2m+1-2j) : 1 \leq j < m\}. \end{aligned}$$

This motivates the question whether, for every integer $m \geq 2$,

$$S_{2m} + 2\beta(2m)\log 2 \in \text{span}_{\mathbb{Q}}(\mathcal{B}_m). \quad (5.13)$$

This is a research question suggested by three weights, not an all-weight theorem. In particular, the fixed coefficient -2 of $\beta(2m)\log 2$ has not been derived in general. The pattern would give a useful upper bound on this family's expression complexity; it would not prove independence of any displayed constants.

The source's separating functional already shows that its S_6 candidate lies outside the restricted single-product presentation. Its exact S_4 proof obtains the missing information from an octahedral involution, a level-four symmetry whose broader role was studied by Zhao [10]. A useful next step is to introduce corresponding convergent weight-seven and weight-nine symmetry relations, then seek short exact certificates for the frozen vectors. The present computation does not supply those certificates.

6 Verification, source audit, and integration notes

6.1 Which computations carry mathematical force?

The article uses three distinct kinds of evidence. The asymptotic and root-perturbation proofs are mathematical arguments with stated parameter ranges. A finite exact certificate completes two arguments by proving rational inequalities on an analytically justified covering or at specified points. Ordinary numerical evaluation is used to diagnose formulas, to illustrate asymptotic orders, and to propose identities. Increasing the precision of that last kind of evidence does not change its logical status.

The compact certificate for Theorem 3.1 is a computer-assisted proof. Its 112 intervals cover the full compact region; monotonicity of the enclosed functions transfers each endpoint calculation to the entire interval. The endpoints outside that region are covered by an analytic estimate and reflection. Thus the proof does not infer a continuous inequality from a finite sample. The certificate contains short rational lower bounds, and the verifier recomputes the stronger rational expressions that imply them.

The nine Lerch sign enclosures play a different role. They prove sign changes at exact positive rational arguments. The pre-existing global zero bound, recalled with its proof mechanism in Section 4, then proves completeness and simplicity of the endpoint zero lists. The continuity and no-escape arguments, rather than numerical continuation, force an interior multiple root.

Finally, the certificates for S_6 and S_8 prove only the inequalities in Proposition 5.3. They are useful mathematical statements about explicitly defined real numbers, but a rational interval of positive width containing zero is compatible with both equality and inequality. No lower bound for a nonzero period combination has been established that would turn these enclosures into a proof of either conjecture.

Verification item	Scope	Role
Harmonic coefficient compiler	Exact symbolic Gaussian moments through d_3 ; pure-gamma normalization at each order	Algebraic cross-check
Uniform harmonic quadrature	24 cases, 75 working decimal digits; small, proportional, transition, large-ratio, and large-shift cases	Numerical diagnostic
Reflected inequality	112 rational intervals, degree-40 series with bounded tails; analytic endpoint coverage	Finite proof certificate
Reflected moment quadrature	Eight proportional and twelve critical-window cases, 65 working digits	Numerical diagnostic
Lerch endpoint signs	Nine rational points, 1000 terms each, 2^{-160} integer grid and monotone tails	Finite proof certificate
Lerch endpoint law and transitions	Nine difference-kernel diagnostics and two ordinary multiple-root candidate solves	Numerical diagnostic
S_6, S_8 residuals	Exact $N = 1100$ and $N = 1200$ Euler enclosures, independent elementary-constant bounds	Proof of proximity
S_8 discovery and audits	Frozen PSLQ vector; fresh 500-digit Euler and independent 220-digit Mellin evaluations	Conjectural evidence

6.2 Diagnostic error orders

The numerical values below use relative error (approximation)/(quadrature) $- 1$. This convention is the reciprocal of the ratio in some theorem statements, and is recorded explicitly in the JSON data. For the harmonic example $p = h$, $a = 1/2$, the first corrected formula gives

h	25	100	400
relative error	$-2.91878 \cdot 10^{-5}$	$-1.92450 \cdot 10^{-6}$	$-1.21909 \cdot 10^{-7}$

Multiplying the depth by four reduces the error by approximately sixteen, consistently with the proved p^{-2} order. At $h = 400$ and $p = h \log h$, the corrected relative error is approximately $3.35 \cdot 10^{-8}$. The 24-case set also includes varying a , very small p/h , and large p/h . These checks are deliberately wider than a single compact-ratio test.

For equal reflected exponents the successive error reductions reported after Corollary 3.3 agree with the n^{-1} and n^{-2} orders. In the critical window the meaningful comparison is against the reference scale $L_{n,m}$, not against the extremely small unnormalized moment. At $n/(m+1) - \log m \simeq 0$, that normalized moment approaches e^d , rather than one.

The numerical programs use scaled integrals and logarithms to prevent the cancellation or underflow inherent in the defining sums. Nevertheless, the quadrature diagnostics are not interval quadrature. Some extreme-case diagnostic errors approach the numerical precision floor, and their last displayed digits carry no proved error guarantee. These cases are omitted from the plotted error comparisons and from judging an empirical power of the remainder.

6.3 Audit of the pinned manuscript

The source is fixed at revision `cc34f73596336f2466d9754cb0f3635bd2bedade`. The provenance manifest records the inspected text files and their content hashes. The audit concentrated on the statements used here and on nearby claims about conductor jets, signed kernels, reflected remainders, and Stieltjes zero geometry. It is not a claim to have independently reproofed every chapter of the book.

The status of S_4 . The current source's `chapters/04-S4-proof.tex` gives an exact octahedral certificate for the mixed weight-five identity. Earlier reports describing that identity as conjectural have been superseded. Accordingly, this article uses S_4 as proved background and does not count it among its new results.

A documentation update for S_6 . The numerical paragraph following the conjecture in `chapters/04-cyclotomic-quotients.tex` correctly describes its 220-digit Mellin and 300-digit Euler values as diagnostics. However, the source's `VALIDATION.md` additionally records a separate 900-term exact integer/fraction replay whose enclosure bounds the *integer-vector* residual by 10^{-260} . The chapter should mention that later certificate so that the narrative and validation ledger express the same evidence level. The proposed replacement paragraph in the accompanying patch keeps the numerical calculations distinct from that certificate and explicitly retains the conjecture status. Our Proposition 5.3 adds a new, separately replayable bound for the *normalized left-minus-right* difference. The normalization must be named whenever residual magnitudes are compared.

The expanding harmonic range. The closing discussion in `chapters/04-proportional-depth.tex` correctly warns that its compact- α theorem does not imply a uniform remainder when $\alpha = p/h$ grows. It asks for one effective formula across that range. Theorem 2.1 now supplies such a formula, and (2.22) gives an explicit answer to the stated question. On integration, the open-problem sentence should point to this theorem, while the warning about the scope of the earlier compact-parameter result should remain.

Fixed reflected exponent versus joint growth. The fixed- m theorems and their sharp remainder statements explicitly hold m fixed. We found no basis for deleting that restriction. The critical-window theorem gives a concrete reason to preserve it: there are sequences with $m/n \rightarrow 0$ for which the normalized leading fixed- m approximation tends to $e^d \neq 1$. This is a limitation of an attempted extension, not a counterexample to the theorem actually stated in the source.

Eventual Lerch saturation versus small derivative order. The source's eventual zero-count theorem takes the parameter derivative order large. Corollary 4.4 instead holds that order equal to one and lets the Lerch parameter approach zero. These are different limits. A sentence asserting the maximal count for every positive deformation at every derivative order would be false, but the inspected theorem does not make that assertion. The new unfolding theorem provides a useful explicit boundary to such a generalization.

No additional substantive mathematical error was identified in the targeted material. The proposed source patch is therefore a correction to the account of available evidence, together with integration guidance for questions now partly or fully answered; it does not announce an unsupported retraction of an existing theorem.

6.4 Reproduction and mathematical dependencies

The package has no network requirement for certificate replay. From its root directory, run

```
python code/replay_exact.py
```

with ordinary Python 3, without optimization that disables assertions. This reconstructs the 112 reflected inequalities, the nine Lerch signs, and both S_6/S_8 rational-enclosure schemes, and compares the results with the stored certificates. The exact proof programs use only the Python standard library. The separate coefficient compiler uses SymPy; numerical diagnostics use mpmath, and the exploratory Lerch coordinate scan uses NumPy and SciPy. Matplotlib reproduces the figures. Tested versions and individual commands are in the README.

The analytic harmonic theorem and the Lerch unfolding and endpoint expansion are independent of numerical diagnostics. The reflected unique-saddle result depends on the finite rational inequality proof; the critical-window theorem has an independent analytic proof. The forced Lerch transitions depend on the nine sign certificates and the proved zero bound. The Gaussian certificate depends on elementary Euler and elementary-constant error bounds, not on the numerical success of PSLQ. None of these programs constitutes a proof of an algebraic relation between evaluated periods unless an exact symbolic relation is separately supplied.

7 Further research questions

The following questions have concrete stopping conditions: an exact identity, a stated uniform remainder, a certified classification, or a counterexample would constitute progress. They are ordered by their proximity to the present results, rather than by a claim that the first questions are easy.

7.1 Gaussian reductions and exact relation certificates

1. Prove or refute the frozen S_6 and S_8 identities. The immediate arithmetic problem is to produce exact symbolic relations for (5.10) and (5.1). The source's separating functional shows why repeating its restricted single-product shuffle calculation cannot suffice for S_6 . A useful approach is to transport the convergent octahedral relations used at weight five to weights seven and nine, and then perform exact elimination. Each analytic relation schema should include its branch, endpoint convergence, and regularization justification before its matrix row is used. The desired output is a short rational linear certificate whose target is the already frozen vector, not another numerical relation search.

2. Find a general even-index reduction mechanism. Does (5.13) hold for every $m \geq 2$? More specifically, can one derive the coefficient -2 of $\beta(2m) \log 2$ before doing weight-specific elimination? An integral transformation that isolates that boundary term could separate the universal part of the problem from the remaining Gaussian double values. A recurrence in m , or a generating function for the entire family, would be stronger than checking additional individual weights. Conversely, an exact obstruction at one weight would identify which new class of periods is required.

3. Compress and certify the relation search. The goal is not just to obtain a matrix of larger rank. Can the octahedral, shuffle, stuffle, conjugation, and distribution relations be organized into a small family of reusable certificates? A valuable deliverable would be a verifier whose trusted analytic inputs are explicitly listed and whose remaining work consists only of rational arithmetic on words. Separating functionals should be retained as diagnostics of missing relation families. Quotient dimensions of a chosen presentation must continue to be distinguished from dimensions of spaces of evaluated periods.

7.2 Uniform asymptotics beyond the present error bounds

4. Make the harmonic uniform constants effective. The proof of Theorem 2.1 establishes universal $C_{J,pJ}$, but does not tabulate useful explicit values. Derive computable majorants for the derivatives of H , choose an explicit localization radius, and propagate these bounds through the Gaussian expansion. The practical target is a relative enclosure for $R_{p,h}(a)$ from the saddle and a fixed number of derivatives, valid without checking which ratio regime applies. Such a result would complement the exact Euler method when the defining alternating sum suffers severe cancellation.

5. Connect the bounded- p and growing- p depth limits. The new theorem allows $p/h \rightarrow 0$ only when $p \rightarrow \infty$. The source has a separate fixed- p large-depth expansion. Can one construct a single approximation with a relative error uniform for p in $[p_0, M(h)]$, including bounded p ? The change from an approximately Gaussian gamma law to a finite-shape gamma law suggests keeping the latter expectation unexpanded until the final step. A successful statement should specify its dependence on a shift a that also varies.

6. Replace the reflected inequality certificate by a conceptual argument. Theorem 3.1 has a complete finite proof, but the positive quantity $D(x)$ deserves a structural explanation. Can it be expressed as an integral with manifestly positive integrand, a covariance of monotone functions, or the derivative of a known gamma-function comparison? Such an argument could give usable lower bounds on the curvature, and might extend to $\log \Gamma(x + \alpha)$ after subtracting an appropriate affine normalization. The affine subtraction and domain must be specified so that the logarithms defining the phase remain real.

7. Obtain one reflected-moment expansion across all exponent ratios. The proportional saddle expansion and the critical-window expectation expansion overlap only after a further analysis. The endpoint saddle has size $e^{-1/\lambda}$, so a hypothesis keeping the saddle away from the boundary is unavailable. Find a change of variable that resolves this exponential motion, then prove a uniform remainder while $n, m \rightarrow \infty$ with no compact restriction on m/n . Reflection should recover the opposite endpoint. The parameter to match is $\tau = me^{-n/(m+1)}$, with the distinction between m and $m+1$ retained at correction order.

8. Classify when the fixed-exponent leading scale remains valid, and extend the transition to all orders. Determine necessary and sufficient conditions, in a stated joint growth domain, for $M_{n,m}/L_{n,m} \rightarrow 1$. The condition $m/n \rightarrow 0$ alone is false, and the critical-window theorem identifies a nonunit limiting profile. It does not by itself settle every sequence outside that

window. For bounded $n/(m+1) - \log m$, derive the general coefficient form

$$\frac{M_{n,m}}{L_{n,m}} \sim e^{d\tau} \sum_{j \geq 0} \frac{P_j(t, \tau)}{m^j},$$

including the degree structure of P_j , a finite coefficient algorithm, and a uniform remainder at each order. The exact gamma moments already used here provide a route to those polynomials; weighted tail control must be kept in the proof.

9. Study late coefficients when the reflected exponent grows. The source's inverse-gamma coefficient analysis and optimal truncation theorems have a separate saddle and a fixed reflected power. Determine how the nearest singularity and least-term remainder change when that power grows with the coefficient index. This is not an automatic consequence of the joint moment theorem: the coefficient-extraction contour and the moment integral have different large parameters and different error budgets.

7.3 Lerch discriminants and real-zero geometry

10. Certify the local transition type at indices three and four. The article proves an interior multiple root but does not prove that it is exactly double. At the displayed numerical candidates, certify a solution of $(D_{n,1}^\rho, \partial_a D_{n,1}^\rho) = (0, 0)$ in a small rectangle and show

$$\partial_a^2 D_{n,1}^\rho \neq 0, \quad \partial_\rho D_{n,1}^\rho \neq 0.$$

These inequalities would establish a transverse local creation or annihilation of two real roots. A separate exclusion argument is needed to prove that this is the only transition in $0 < \rho < 1$. The endpoint logarithmic law helps determine a suitable parameter scale, but its simple-root formula cannot be used at the multiple root itself.

11. Determine the zero-count diagram for arbitrary indices and derivative orders. For $n > k$, the general unfolding theorem reduces the small-parameter count to the sign of $P_{n,k}(-\log 2)$, with its nonvanishing stated as a hypothesis. Can one determine that sign by elementary inequalities over substantial ranges of n, k ? Can the endpoint n -zero count be proved uniformly in larger families without separate numerical point selection? Together these results would give lower bounds on the number of interior discriminant events. They would also quantify the derivative-order threshold beyond which the source's eventual saturation holds for all ρ .

12. Resolve the full endpoint expansion and its interaction with zeros. Theorem 4.5 isolates the leading $\epsilon^k L^{n+1}$ term. Determine the complete polynomial in $L = \log(1/\epsilon)$ at order ϵ^k , and continue to higher powers of ϵ with explicit dependence on a . One expects the lower logarithmic coefficients to record the shift and the differentiated logarithmic polynomial that the leading coefficient forgets. A Mellin-transform derivation should be compared with the elementary split-sum proof, with all contour shifts and multiple-pole contributions justified. The resulting expansion could give sharper trajectories of simple zeros and a starting point for simultaneous limits in which the Stieltjes index also grows.

The research programme above connects analytic structure, finite exact proof objects, and high-precision discovery. Each question specifies which part of that work remains open and what would constitute a checkable advance.

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